

1. Composition and the chain rule

For two functions f and g , their composition is

$$(f \circ g)(x) = f(g(x)).$$

Its domain consists of the points x in the domain of g for which $g(x)$ lies in the domain of f . For a product, the known rule is

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x).$$

We now need a rule for a composition.

Example

Let $f(u) = u^2$ and $g(x) = 3x + 1$. Then

$$f(g(x)) = (3x + 1)^2 = 9x^2 + 6x + 1,$$

so its derivative, found by expanding, is $18x + 6$. The same answer can be written as

$$f'(g(x))g'(x) = 2(3x + 1) \cdot 3.$$

The derivative of f is evaluated at $g(x)$, the argument that is supplied to f .

Theorem (chain rule). Let I, J be open intervals, let $g : I \rightarrow J$, and let $f : J \rightarrow \mathbb{R}$. Fix $a \in I$. If g is differentiable at a and f is differentiable at $g(a)$, then $f \circ g$ is differentiable at a , and

$$(f \circ g)'(a) = f'(g(a))g'(a).$$

Only the behavior near a and near $g(a)$ matters. In applications, it is enough to choose suitable small intervals on which the composition is defined.

Notation. If $u = g(x)$ and $y = f(u)$, the rule is often written

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}.$$

Here $dy/du = f'(u)$ is evaluated at $u = g(x)$. This notation records the chain-rule identity.

Example

Suppose $g(2) = 3$, $g'(2) = -1$ and $f'(3) = 4$, with the differentiability and domain hypotheses of the theorem satisfied. Then

$$(f \circ g)'(2) = f'(3)g'(2) = 4(-1) = -4.$$

No formula for the two functions is needed.

2. Proof of the chain rule

Write $b = g(a)$. Define a function on J by

$$Q(u) = \begin{cases} \frac{f(u) - f(b)}{u - b}, & u \neq b, \\ f'(b), & u = b. \end{cases}$$

The definition of $f'(b)$ says exactly that

$$\lim_{u \rightarrow b} Q(u) = f'(b) = Q(b).$$

Thus Q is continuous at b . Its defining identity is

$$f(u) - f(b) = Q(u)(u - b) \quad (u \in J).$$

This identity also holds at $u = b$, since both sides are zero.

Substituting $u = g(a + h)$, for nonzero h sufficiently close to zero we have

$$\frac{f(g(a + h)) - f(g(a))}{h} = Q(g(a + h)) \frac{g(a + h) - g(a)}{h}.$$

Since g is differentiable at a , it is continuous there. Continuity of Q at $b = g(a)$, followed by continuity of composition, gives

$$Q(g(a + h)) \longrightarrow Q(g(a)) = f'(g(a)).$$

The other factor tends to $g'(a)$. By the product limit law,

$$\lim_{h \rightarrow 0} \frac{f(g(a + h)) - f(g(a))}{h} = f'(g(a)) g'(a).$$

This proves the theorem. $\square$

The function Q avoids dividing by $g(a + h) - g(a)$, which may be zero even when $h \neq 0$. For example, a constant inner function gives this situation for every h . The argument above still applies, and the derivative of the composition is zero.

Example

The hypotheses are sufficient, but need not hold whenever a composition is differentiable. Let $f(u) = |u|$, $g(x) = x^2$. Then

$$(f \circ g)(x) = |x^2| = x^2$$

is differentiable at zero, although f is not differentiable at $g(0) = 0$. Its derivative at zero follows from the simplified formula; the chain-rule theorem does not apply there.

3. Applications of the chain rule

If $u = u(x)$ is differentiable, the known formulas and the chain rule give

$$\begin{aligned}\frac{d}{dx}[u(x)]^n &= n[u(x)]^{n-1}u'(x), \\ \frac{d}{dx}\sin(u(x)) &= \cos(u(x))u'(x), \\ \frac{d}{dx}\cos(u(x)) &= -\sin(u(x))u'(x), \\ \frac{d}{dx}e^{u(x)} &= e^{u(x)}u'(x), \\ \frac{d}{dx}a^{u(x)} &= a^{u(x)}\ln(a)u'(x) \quad (a > 0).\end{aligned}$$

Here the power formula is for a nonzero integer n ; if $n < 0$, require $u(x) \neq 0$. For $n = 1$, it is simply $u'(x)$. The formulas apply wherever the stated differentiability and domain conditions hold.

Example

For $H(x) = (1 + x^2)^{-3}$, take $u(x) = 1 + x^2$. Then

$$H'(x) = -3(1 + x^2)^{-4}(2x) = \boxed{-\frac{6x}{(1 + x^2)^4}}.$$

Since $1 + x^2 > 0$ for every real x , both H and H' are defined on $\mathbb{R}$.

Three functions. Applying the chain rule twice gives

$$\frac{d}{dx}f(g(h(x))) = f'(g(h(x)))g'(h(x))h'(x),$$

at points where the corresponding differentiability and domain hypotheses hold.

Example

Let $H(x) = e^{\sin(x^2)}$. The successive values are x^2 , then $\sin(x^2)$, then $e^{\sin(x^2)}$. Therefore

$$H'(x) = e^{\sin(x^2)}\cos(x^2)(2x) = \boxed{2x\cos(x^2)e^{\sin(x^2)}}.$$

All functions involved are differentiable on $\mathbb{R}$.

Example

Let $P(x) = xe^{x^2}$. Apply the product rule, using the chain rule for the second factor:

$$P'(x) = e^{x^2} + x(e^{x^2}(2x)) = \boxed{e^{x^2}(1 + 2x^2)}.$$

4. Implicit differentiation

An equation involving x and y need not specify a single function y of x . For example,

$$x^2 + y^2 = 25$$

describes a circle. For $-5 < x < 5$, its upper and lower halves are given separately by

$$y = \sqrt{25 - x^2} \quad \text{and} \quad y = -\sqrt{25 - x^2}.$$

A **local branch** $y = y(x)$ means one function describing part of the curve on an interval of x -values.

Suppose such a branch is differentiable and satisfies the equation throughout an open interval. We may differentiate the two equal functions of x . The occurrence of y means $y(x)$, so the chain and product rules give, for example,

$$\frac{d}{dx}(y^2) = 2y y', \quad \frac{d}{dx}(xy) = y + xy'.$$

Similarly, $\frac{d}{dx} \sin y = \cos y y'$ and $\frac{d}{dx} e^y = e^y y'$. Here $y' = dy/dx$.

Example

For a differentiable branch of $x^2 + y^2 = 25$,

$$2x + 2y y' = 0.$$

At a point where $y \neq 0$, we can solve:

$$\boxed{y' = -\frac{x}{y}}.$$

At $(3, 4)$ the slope is $-3/4$, whereas at $(3, -4)$ it is $3/4$. The point on the curve determines which value of y to use.

Why the branch assumption is valid here. Both functions $y_{\pm}(x) = \pm\sqrt{25 - x^2}$ are continuous for $-5 < x < 5$, by root and composition continuity. Fix a in this interval. For either branch, rationalization gives

$$\frac{y_{\pm}(a+h) - y_{\pm}(a)}{h} = -\frac{2a+h}{y_{\pm}(a+h) + y_{\pm}(a)} \rightarrow -\frac{a}{y_{\pm}(a)}.$$

The limiting denominator is $2y_{\pm}(a) \neq 0$. This proves differentiability directly and verifies the formula.

5. An implicit curve and its tangent

Example

Consider the curve

$$x^3 + y^3 = 6xy.$$

The point $(3, 3)$ lies on it, since $27 + 27 = 6 \cdot 3 \cdot 3$. For this example, we accept that near $(3, 3)$ the curve has a differentiable branch $y = y(x)$. We will calculate its derivative without solving the equation for y .

Differentiate the identity along this branch:

$$3x^2 + 3y^2y' = 6(y + xy').$$

Collect the terms containing y' :

$$(3y^2 - 6x)y' = 6y - 3x^2.$$

Where $y^2 - 2x \neq 0$, this gives

$$y' = \frac{2y - x^2}{y^2 - 2x}.$$

At $(3, 3)$, the denominator is $9 - 6 = 3$, so

$$y'(3) = \frac{6 - 9}{9 - 6} = -1.$$

The tangent line is therefore

$$y - 3 = -(x - 3), \quad \text{or} \quad x + y = 6.$$

The steps used.

1. Check that the point satisfies the original equation.
2. Regard y as a function of x and differentiate the identity, using the chain rule whenever $y(x)$ occurs inside another function.
3. Collect the terms containing y' and solve where their coefficient is nonzero.
4. Evaluate at the specified point and use $y - y_0 = m(x - x_0)$ for the tangent.

The derivative formula has been derived for differentiable branches where its denominator is nonzero. At a point where the denominator vanishes, we return to the differentiated equation or examine the curve more directly.

6. Domains and vertical tangents

A finite slope. At a point (x_0, y_0) on a differentiable branch, a finite derivative m gives the tangent

$$y - y_0 = m(x - x_0).$$

For example, the circle at $(3, 4)$ has tangent

$$y - 4 = -\frac{3}{4}(x - 3), \quad \text{equivalently} \quad 3x + 4y = 25.$$

A vertical tangent. At $(5, 0)$, the circle formula $y' = -x/y$ cannot be used. Indeed, any differentiable branch $y = y(x)$ through that point would give $10 + 0 \cdot y' = 0$, which is impossible.

Instead describe the right side of the circle as a function of y :

$$x(y) = \sqrt{25 - y^2}, \quad -5 < y < 5.$$

At $y = 0$, the difference quotient is

$$\frac{x(t) - x(0)}{t} = \frac{\sqrt{25 - t^2} - 5}{t} = -\frac{t}{\sqrt{25 - t^2} + 5} \rightarrow 0.$$

Thus $dx/dy = 0$ there. Writing the tangent formula with y as the independent variable gives

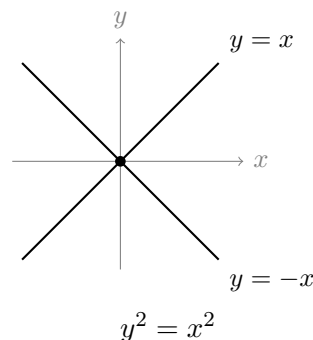
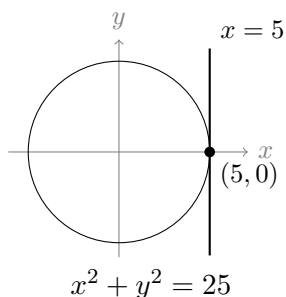
$$x - 5 = 0(y - 0), \quad \boxed{x = 5}.$$

The left side similarly has vertical tangent $x = -5$ at $(-5, 0)$. A vertical tangent is not a finite value of dy/dx .

A crossing. Now consider

$$y^2 = x^2.$$

This curve is the union of the two lines $y = x$ and $y = -x$. Implicit differentiation gives $2y y' = 2x$. At $(0, 0)$, that equation becomes $0 = 0$ and supplies no slope. The two branches have slopes 1 and -1 , respectively, and two distinct tangent lines at the same point. The undefined expression x/y at the origin therefore does not establish a vertical tangent.



7. Derivatives of inverse functions

Let $g = f^{-1}$. The inverse identities are

$$f(g(x)) = x, \quad g(f(t)) = t.$$

If g is differentiable, the chain rule applied to the first identity gives

$$f'(g(x)) g'(x) = 1.$$

Thus, wherever the hypotheses hold,

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}.$$

The derivative in the denominator is evaluated at $f^{-1}(x)$. The notation f^{-1} denotes an inverse function, not a reciprocal.

Theorem. Let f be continuous and strictly monotone on an open interval I , and let $g = f^{-1}$ on $J = f(I)$. If f is differentiable at $a \in I$ and $f'(a) \neq 0$, then g is differentiable at $b = f(a)$, with

$$g'(b) = \frac{1}{f'(a)}.$$

The continuity and monotonicity assumptions make J an open interval and ensure that the inverse is defined there.

Justification. First, g is continuous at b . To see this when f is increasing, choose any sufficiently small $\eta > 0$ with $a - \eta, a + \eta \in I$. Then

$$f(a - \eta) < b < f(a + \eta).$$

If x is sufficiently close to b , it lies between these two bounds; strict monotonicity then gives $a - \eta < g(x) < a + \eta$. The same argument with reversed inequalities applies when f is decreasing. Thus $g(x) \rightarrow a$ as $x \rightarrow b$.

For $x \neq b$, put $t = g(x)$. Then $t \neq a$, $x = f(t)$, and

$$\frac{g(x) - g(b)}{x - b} = \frac{t - a}{f(t) - f(a)} = \frac{1}{\frac{f(t) - f(a)}{t - a}}.$$

As $x \rightarrow b$, we have $t \rightarrow a$, so the denominator tends to $f'(a) \neq 0$. The reciprocal limit law proves that $g'(b) = 1/f'(a)$. $\square$

This argument establishes differentiability of the inverse; the chain-rule calculation above then expresses the result using the inverse identity.

8. Logarithm and arcsine

Example

The function $f(t) = e^t$ is continuous and strictly increasing on $\mathbb{R}$, with range $(0, \infty)$, and $f'(t) = e^t > 0$. Its inverse is $g(x) = \ln x$. The inverse derivative formula gives

$$\boxed{\frac{d}{dx} \ln x = \frac{1}{e^{\ln x}} = \frac{1}{x}, \quad x > 0.}$$

The restriction $x > 0$ is the domain of the logarithm. The reciprocal expression $1/x$ is also defined for negative x , but it is not the derivative of $\ln x$ there, since $\ln x$ is not defined there.

Equivalently, write $y = \ln x$, so $e^y = x$. Differentiating gives $e^y y' = 1$, hence $y' = 1/e^y = 1/x$. The theorem justifies the differentiability used in this calculation.

Example

The inverse sine is defined by restricting sine to $[-\pi/2, \pi/2]$. Thus

$$y = \arcsin x \iff \sin y = x, \quad -\pi/2 \leq y \leq \pi/2.$$

For $-1 < x < 1$, we have $-\pi/2 < y < \pi/2$. On this open interval sine is continuous and strictly increasing, and its derivative satisfies $\cos y > 0$. The inverse derivative theorem therefore applies and gives

$$\frac{d}{dx} \arcsin x = \frac{1}{\cos(\arcsin x)}.$$

Using $\sin y = x$ and the identity $\sin^2 y + \cos^2 y = 1$, we obtain

$$\cos y = \sqrt{1 - \sin^2 y} = \sqrt{1 - x^2}.$$

The positive square root is required because $\cos y > 0$ on the chosen interval. Consequently,

$$\boxed{\frac{d}{dx} \arcsin x = \frac{1}{\sqrt{1 - x^2}}, \quad -1 < x < 1.}$$

The function $\arcsin$ has domain $[-1, 1]$; its derivative has domain $(-1, 1)$. At the endpoints the denominator vanishes, and there is no finite two-sided derivative under our definition.